

# Guidelines for writing scientific publications - Simple rules to follow for reports, theses and papers

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2018.03.13.

## Overview

This note is meant for BSc, MSc and PhD students, with the purpose of improving the quality, understandability and readability of reports, studies, theses and papers. It gives a quick start on how to prepare a scientific document, highlighting some basic, yet often overlooked principles that should be adhered to. Hopefully adopting these simple rules will increase the quality of scientific documents (primarily written by students) and streamline the writing and reviewing process.

Section 1 gives a list of the bare minimum rules that must be respected in any scientific document, whereas Sections 2-4 cover specific aspects in more details. The best is to read the whole document before you write a scientific document for the first time, look up Section 1 for refreshing your memory the first few times you write, and look up specific parts of Sections 2-4 when needed.

## 1 Quick start for the impatient

- **Always use exact references** ("Equation 1 shows", "in Figure 1 we see", etc.).
- **Never use non-specific references** (e.g. "the above Equation", "earlier we saw").
- **Introduce all abbreviations at their first use.**
- **Introduce all mathematical variables at their first use.**
- **Mathematical notation must be unique**, never use the same notation for different variables, functions, etc.
- **Figures and tables must be readable and easy to interpret on their own.** Include longer captions to explain the main message of the figure/table, so that they can practically be read without the manuscript.
- **Use proper English, with appropriate tenses and phrases.** Check your text with word processing programs for grammar mistakes and typos.
- **Be exact, concise, and professional.** Do not use contractions, profanity or colloquialism.
- **Structure your work with an explicit message in the focus.** Everything you write should support your message and highlight its context.
- **Google Is Your Friend. Use it.**

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## 2 Know what you want to communicate to whom

You are writing a scientific document, because you have done research and you want to communicate your work to others. The key to successfully doing so is to be clear, concise and easy to understand. No matter how brilliant your work is, if you cannot explain it to your intended audience with words, they will not care about what you have done.

- **Your prime directive should always be a clear message.** You are telling others about something very specific you have done with an outcome that has scientific value. Thus, your work must have an explicit message and this message should be clear throughout your document, from start to finish. Everything you write is there to support your message and put it in proper context.
- **Know your readers.** The level of necessary background information, technical details and experimental methodology description that you need to include depends on who will read your work. A thesis is read by people with a general knowledge of the field and needs more details, whereas a scientific journal is mostly read by experts and you can assume a higher familiarity with the basics.
- **Start writing at the end, by summarizing your results and their consequences.** Once you have done so, you know exactly the message you want to deliver and you can build up the rest of the document in its support.
- **Structure your work.** Your document should be composed of a few sections (2-5), each targeting a broad aspect of the research (its background, the methodology you used, the results you achieved, etc.). Your sections should be made up by a few (2-5) subsections, focusing on different parts of the broader topic being discussed. Your smallest structural units are the paragraphs, each representing a central, specific thought, together containing all details relevant to the given subsection.
- **Think at the paragraph level, expressing one concept at a time.** A paragraph should be made up by a few sentences (3-10 is a practical range), with the first sentence summarizing the specific message of the paragraph, and the rest giving all background information. Use conjunctions and conjunctive adverbs to ensure coherence and logic, both within the paragraphs and between them (see Section 4.3 for a detailed list of phrases).
- **Write clearly and concisely.** Make sure that each part of your text is understandable and gives all details needed to reproduce your work, but do not write more than as necessary.
- **Do not be afraid of longer, more descriptive section titles.** Some journals enforce a strict overall structure (e.g. "Introduction", "Methods and Materials", "Results", "Discussion" and "Conclusion"), but even in such cases you have freedom over smaller structural units. Reading through only your chapter, section and subsection titles should give the reader a good general idea of what you are reporting on.
- **"Fuzzy writing reflects fuzzy thinking."** If you cannot clearly translate your thoughts, ideas and the work you did into words, chances are you do not understand what you have done as well as you think you do.

## 3 Avoid basic mistakes confusing the readers

The most important aspects of a scientific document are its clarity, understandability and reproducibility. While the specifics of how to write in a simple, yet clear way may vary between disciplines, there are several common, basic rules that are universally applicable and make any report better. Following these ensures that the reader at least has a chance to understand your work.

### 3.1 Always be explicit when referencing

- **All references must be exact and unambiguous.** Use references such as "Equation 1 shows", "in Figure 1 we can see", "Table I summarizes", "Section 2 introduced", "John Doe [1] showed that").

- **It is never acceptable to use non-specific references.** Never use "the above Equation", "as shown in the Figure below", "earlier we saw", "the variable in the inequality", etc.
- **Referring with "this", "that", "which", "who", "these", etc. is only allowed if they unambiguously relate to what has just been said.** This typically means they can only refer to the immediately preceding sentence or the previous part of a sentence. As a general rule, try to minimize or eliminate their use.

### 3.2 Use abbreviations and acronyms clearly and in moderation

- **All abbreviations must be introduced at the point they are first used.** Never use an abbreviation without introducing it, no matter how widespread its use may be.
- **Avoid the excessive use of abbreviations.** It may sound clever to "calculate the TCP for the CTV using an EUD and LKB based model with a BED component", but it is bad writing. Always aim for maximal clarity and understandability.
- **Avoid using abbreviations in figures, tables and their captions.** Figures and tables should practically be self-explanatory even for someone who does not read any other part of your document.
- **Avoid using abbreviations in titles,** unless absolutely necessary. It is hardly ever necessary.
- **Use Latin abbreviations and phrases with their correct meaning.** Do not confuse "i.e." with "e.g.", "in vitro" with "in vivo", etc. Write "et al." correctly without a period after "et", append a comma after "e.g.", and use "that is" and "for example" outside parentheses.

### 3.3 Typeset mathematical formulas consistently

- **All mathematical variables must be introduced at the point they are first used.**
- **Avoid using variables excessively in the main text,** typically it is better to use them together with their definition (e.g. write "as Equation 1 shows the acceleration  $\vec{a}$  depends on the force  $\vec{F}$  and the mass  $m$ ", rather than "Equation 1 shows that  $\vec{a}$  depends on  $\vec{F}$  and  $m$ ").

$$\vec{F} = m\vec{a} \tag{1}$$

- **The mathematical notation must be uniquely defined and unambiguous.** It is never allowed to use the same notation for different variables, functions, sets, etc. in the same document (or at least within the same chapter). For example, if you defined  $\vec{F}$  as the force vector, you cannot use  $\vec{F}$  to describe a discretized flow field.
- **Your notation must be uniform.** Choose a style to denote vectors, matrices, sets, etc. and stick to it throughout the document.
- **All physical quantities must include their units,** numbers without units are meaningless. Use standard SI units. Hardly any circumstance justifies using English or CGS units.
- **You must be able to describe any and all numerical methods you use or develop in exact, well defined mathematical terms.** Describing an algorithm vaguely in words alone is insufficient, it has to be backed up with the proper, clear mathematical description of every step. If you have difficulty explaining with formulas what your method/algorithm/program does, you probably do not understand what you are doing as well as you think you do.

### 3.4 Prepare your tables and figures with the utmost attention

- **Figures and tables are the most important part of your work.** The main message you want to communicate should be clear even to someone who only looks at the figures and tables, reading their captions, but reading nothing from the main text. High quality, self-explanatory, shiny and pretty figures are always the way to the reader's heart.

- **All figures need to be readable and easy to interpret.** Avoid putting too many lines and markers in one figure. All font sizes (in caption, labels, tick labels, legends and text boxes) must be large enough to be easily read.
- **All tables need to be readable, easy to interpret and justified.** Tables are tools for presenting large amounts of data necessary for properly interpreting your work and ensuring its reproducibility. They are not page fillers for beefing up your document.
- **Do not be afraid of longer captions** to explain the essence of what figures and tables present.

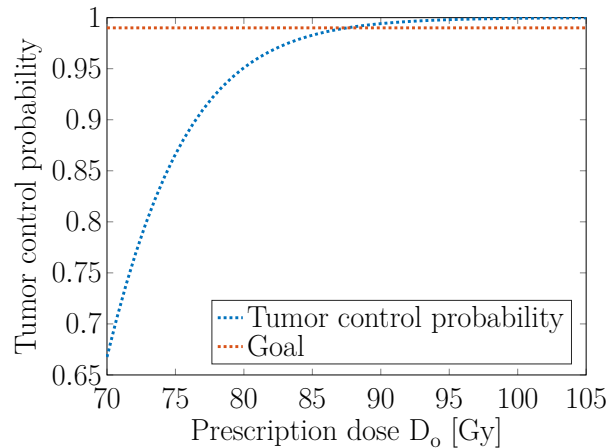


Figure 1. An example of a good figure showing the dependence of the tumor control probability on the prescription dose  $D_0$  in a tumor with a radioresistant cell sub-population.

## 4 Write in your own voice, but always aim for clarity

As the phrase goes, "style is the man himself". Your document reflects you, your personality and your style. You should write it with your own tone and should not try to mimic someone else's manners. Nevertheless, you are writing a scientific document where clarity is key, and following a few stylistic suggestions and conforming to standard English go a long way in helping you achieve that.

### 4.1 Use tenses appropriately

There are 12 English tenses, each suitable under specific circumstances. However, for most situations simple past, simple present, simple future and present perfect should be sufficient, together with passive voice and conditional sentences when necessary.

- **Use the simple past for describing events that happened in the past and have finished.** For example, the work others did in the field is in the past ("John Doe [1] et al. showed that ..."). The work you did and are currently reporting on is also in the past ("our cell cultures were kept at room temperature").
- **Use the simple present to make general statements.** For example, mathematical definitions and method descriptions all fall into this category (" $m$  is the mass of the apple,  $\vec{F}$  is the gravitational force and  $\vec{a}$  is the resulting acceleration.", "Equation 1 describes Newton's second law.").
- **Use the simple future for events that will happen in the future.** E.g., "Our follow up study will focus on something even more interesting".
- **Use the present perfect for things that happened in the past but have a consequence in the present** ("Newton's work has fundamentally changed our view of the universe.").

- **Use conditional sentences to describe hypothetical situations and implications.** The three typical situations apply to the future ("If we invent a faster than light (FTL) technology in the future, we will be able to colonize the universe"), the present ("If we had FTL technology today, we would be able to colonize the universe.") and the past respectively ("If our ancestors had had FTL technology, they would have been able to colonize the universe.").

## 4.2 Use professional tone without colloquialism

Your readers are scientists, not your buddies. Your tone must reflect this and your text should read professional. Better to be dry, factual and clear than emotional and vague. Profanity, cursing or contractions are never allowed. You can note that "Unfortunately the homework assignment did not specify the mass of the apple." or that "The difference between general relativity and Newton's gravity law is only 1% for this case", but you can never write that "The stupid description didn't tell me the mass of the apple." or that "I think general relativity is useless and way too complicated for this problem.".

## 4.3 Use conjunctions and conjunctive adverbs to express your logic

There are a vast number of conjunctions and conjunctive adverbs in English. They help making logical connections between sentences and parts of a sentence, but only when used in their correct context. The most important conjunctions and conjunctive adverbs, with their meaning and example sentences are summarized in Table I.

## 4.4 If in doubt, always remember, Google Is Your Friend (GIYF)

See <https://www.youtube.com/watch?v=ftQ6A3DKKeg> for never ever forgetting this.

## References

- [1 ] John Doe, et al. "The theory of writing good scientific stuff". *Imaginary Journal of Scientific Publishing*, 1985.

Table I. The most important conjunctions and conjunctive adverbs, their meaning and example uses.

| Conjunction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>although, even though, though</b><br><b>but, yet</b><br><b>as, since, because</b><br><b>even if</b><br><b>whenever</b><br><b>wherever</b><br><b>either ... or</b><br><b>not only ... but also</b><br><b>neither ... nor</b><br><b>both ... and</b><br><b>whether ... or</b>                                                                                                                                                                                                                                                                                           | despite the fact that<br>exception or contrast<br>for the reason that<br>even in the case that<br>at whatever time<br>at whatever place, in every case<br>expresses alternative<br>expresses parallelism<br>a link with negative connotation<br>a link with a positive connotation<br>expresses alternative                                                                                                                                                            | Though gravity is a fundamental force, it is insignificant on sub-atomic levels.<br>Newton was a brilliant, but vindictive person.<br>The apple fell from the tree, since Earth's gravity pulled it down.<br>Even if you learn, you can still fail an exam.<br>Google can help you whenever you need it.<br>Wherever possible, try googling first.<br>You can demonstrate gravity either with falling apples or with falling pears.<br>Newton was not only brilliant at physics, but also at mathematics.<br>Neither Newton, nor Galilei knew about relativity theory.<br>Both Galilei and Newton were fundamental to advancing science.<br>You can vote on whether Newton or Galilei was more brilliant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Conjunctive adverbs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>accordingly</b><br><b>additionally, furthermore, moreover</b><br><b>alternatively</b><br><b>comparatively</b><br><br><b>consequently, hence, therefore, thus</b><br><br><b>conversely, in contrast</b><br><br><b>equally</b><br><br><b>finally</b><br><b>instead, rather</b><br><br><b>analogously, correspondingly, likewise,</b><br><b>equivalently, similarly</b><br><b>meanwhile, concurrently, in the</b><br><b>mean time, simultaneously</b><br><b>namely, specifically</b><br><br><b>nevertheless, nonetheless, however</b><br><b>afterwards, subsequently</b> | in agreement with<br>in addition to<br>as a substitute/another possibility<br>to a moderate degree in comparison<br>to something<br>as a result of<br><br>reversing what was just said<br><br>to the same extent/in the same<br>manner to<br>after a long time/in the end<br>as an alternative<br><br>in a similar, not too different manner<br><br>at the same time/in the intervening<br>period<br>that is to say<br><br>in spite of<br>after something has happened | Example<br>Newton was brilliant. Accordingly, he fundamentally changed science.<br>Newton was a scientist. Additionally, he was a warden of the Royal Mint.<br>Use Newton's law of gravity. Alternatively, use general relativity theory.<br>Discovering gravity fundamentally changed science. Relativity theory had a comparatively important effect.<br>The gravitational force is proportional to objects' masses. Consequently, heavy black holes have very strong gravitational fields.<br>Gravity has an infinite range. Conversely, the strong interaction is practically unobservable at distances larger than 10 fm.<br>Gravity has infinite range. The electromagnetic force has an equally infinite range.<br>Finally, we conclude that this note is very helpful.<br>Newton's law of gravity is inaccurate in strong gravitational fields. Instead, use the laws of general relativity in such situations.<br>Discovering gravity was hugely important. Discovering relativity theory had a similarly large impact.<br>The tested cells were cooled to 0°C. Meanwhile, the control cells were kept at room temperature.<br>Some fundamental forces, namely gravitational and electromagnetic interactions, have long range effects.<br>Newton was a brilliant scientist. Nevertheless, he was an unpleasant person.<br>Newton described the gravitational force in 1687. Subsequently, it became possible to predict celestial movements. |